



2025

COMPUTER SCIENCE — GENERAL

Paper : SEC-B-1 and SEC-B-2

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

SEC-B-1

(Multimedia and its Applications)

Full Marks : 80

Answer **question nos. 1 and 2** and **any four** questions from the rest.

1. Answer *any ten* questions :

2×10

- (a) What is vector graphics?
- (b) What is lossless compression?
- (c) Name any two popular video codecs.
- (d) Define the term bandwidth in multimedia networking.
- (e) What is Virtual Reality Modeling Language (VRML)?
- (f) What is frame rate in video?
- (g) Name two open-source image editing software.
- (h) What is morphing in animation?
- (i) What is anti-aliasing?
- (j) What is GIF used for?
- (k) Define aspect ratio with example.
- (l) What is chroma key?
- (m) What is subtitle encoding?
- (n) What are NTSC and PAL?
- (o) What is multimedia authoring software?

2. Write short notes on (*any four*) :

5×4

- (a) CODEC
- (b) Use of multimedia in education
- (c) Audio sampling techniques
- (d) QoS parameters
- (e) MIDI
- (f) Hypertext and its applications.

Please Turn Over

(2035+2036)

3. (a) Define Digital Audio. How is it created and stored in computers?
(b) What is OCR? Explain its advantages. 5+5
4. (a) Explain various video file formats commonly used in multimedia.
(b) What is video transcoding? Why is it important? 6+4
5. (a) Differentiate between Raster and Vector images with example.
(b) What is Dithering? Explain its significance. 6+4
6. (a) What is VOIP? Explain its importance in multimedia conferencing.
(b) How is point-to-point video conferencing different from multipoint video conferencing? 5+5
7. (a) Describe the key components of a Multimedia Application.
(b) What is color depth? Explain with suitable example. 6+4
8. (a) Discuss different types of image compression techniques.
(b) Write a brief note on additive and subtractive color models. 5+5